# Curvature Flow

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## Flow on a Disk

The two standard real coordinate systems on $\mathbb{R}^2$ are $(x, y)$ and $(r, \theta)$, and there is also a single complex coordinate given by $z = x + iy$, and another given by $\bar{z} = x - iy$. These are related as follows:

$$(r, \theta) = (\sqrt{x^2 + y^2}, \arctan(y/x)), \qquad (x, y) = (r\cos\theta, r\sin\theta),$$

$$z = re^{i\theta}, \qquad \bar{z} = re^{-i\theta}.$$

The operators $\partial_x = \frac{\partial}{\partial x}$, etc. can be thought of as vectors, so that $\partial_x f$ represents the change in $f$ in the direction of the vector $\partial_x$. Using some differential geometry, we have the following relationships:

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y),$$

$$\partial_z = \frac{e^{-i\theta}}{2r}(r\partial_r - i\partial_\theta), \qquad \partial_{\bar{z}} = \frac{e^{i\theta}}{2r}(r\partial_r + i\partial_\theta).$$

The standard laplacian is $\Delta = \partial_{xx} + \partial_{yy}$. By composing the above formulas and (using the product rule), we can also write

$$\Delta = 4\partial_{\bar{z}}\partial_z, \tag{1}$$

$$= \partial_{rr} + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_{\theta\theta}. \tag{2}$$

Based on our experience so far, it seems like polar coordinates are best for curvature flow and complex coordinates are best for calculating geodesics.

The *standard metric* $g_0$ can be expressed as $g_0 = (\mathrm{d}x)^2 + (\mathrm{d}y)^2$ or $g_0 = (\mathrm{d}r)^2 + r^2(\mathrm{d}\theta)^2$ or $g_0 = |\mathrm{d}z|^2$. This is technically called a *symmetric* $(0, 2)$*-tensor*, but as far as we are concerned it simply represents the method being used to measure the (squared) length of vectors.

Associated to $g_0$ is the *standard symplectic form* $\omega_0 = \mathrm{d}x \wedge \mathrm{d}y$ or $\omega_0 = \frac{1}{2i}\mathrm{d}\bar{z} \wedge \mathrm{d}z$ or $\omega_0 = r\,\mathrm{d}r \wedge \mathrm{d}\theta$, which is an *antisymmetric* $(0, 2)$*-tensor* used to measure the area of parallelograms. These are mostly used for integration.

As in the stereographic projection example, we often want to use portions of $\mathbb{R}^2$ to represent patches of a surface which has different geometry. This makes the notion of distances between points vary depending on the location in $\mathbb{R}^2$. Thus, we can used a positive *weight function* $u$ on $\mathbb{R}^2$ to

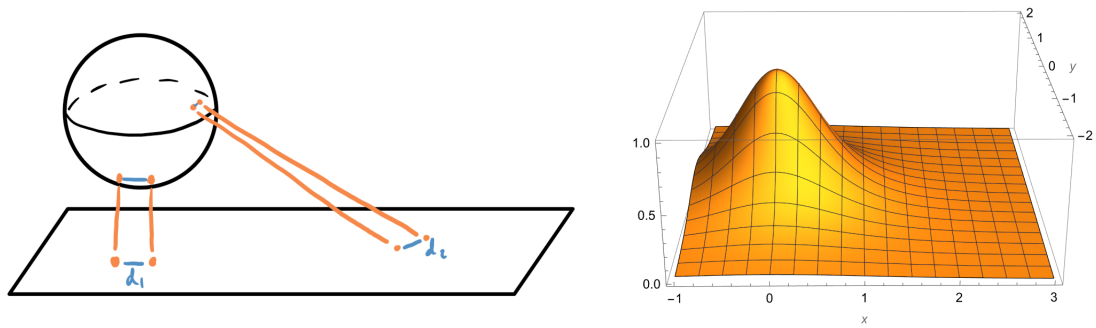


Figure 1: The image on the left is the stereographic projection, wherein the metric should measure $d_1$ as larger than $d_2$. Thus, the weight function $u$ should be smaller away from the origin in $\mathbb{R}^2$. This is reflected in the image on the right, which shows the graph of the weight function for the Fubini-Study metric.

create a new metric $g = ug_0$ (see Figure 1). In the various coordinate systems this looks like[1]

$$g = u(x,y)\,(\mathrm{d}x^2 + \mathrm{d}y^2) \tag{3}$$
$$= u(r,\theta)\,(\mathrm{d}r^2 + r^2\mathrm{d}\theta^2) \tag{4}$$
$$= \frac{1}{2i}u(z)\,|\mathrm{d}z|^2. \tag{5}$$

The associated symplectic form is $\omega = u\,\omega_0$. When doing curvature flow, this weight function $u$ is what evolves over time. Some, such as Jost, use $\lambda^2$ in place of $u$ to avoid worrying about negative values. However Mazzeo-Rubinstein-Sesum just use $u$.

**Definition.** Given a metric $g = ug_0$, the *metric laplacian* is $\Delta_g = \frac{1}{u}\Delta$ where $\Delta$ is the standard laplacian, and the *scalar curvature* is $K_g = \Delta_g \log u$.

The reason for the $\frac{1}{u}$ factor has to do with coordinate changes. To illustrate this, consider the act of "zooming in" to a point on the sphere, which makes the curvature appear 'flatter.' The formula $\Delta \log u$ will reflect this fact, which is undesirable because we want the curvature to be a fixed number independent of what coordinates we are using. Using $K_g = \Delta_g \log u = \frac{1}{u}\Delta \log u$ accounts for this.

Note that $K_g$ is a function on $\mathbb{R}^2$, because the curvature depends on the location. The purpose of Ricci flow is to even out the curvature, to make $K_g$ a constant function. The limit is called a *uniformization metric*.

Let $g_t = u_t g_0$ be a family of metrics[2], and let $K_t = K_{g_t}$. Let $U \subseteq \mathbb{R}^2$ be a region. With respect to the metric $g_t$, the volume (area) of $U$ is $\mathrm{Vol}_t(U) = \int_U \omega_t = \int_U u_t \,\mathrm{d}x\,\mathrm{d}y$ and the average curvature on $U$ is

$$\rho_t = \frac{1}{\mathrm{Vol}_t(U)} \int_U K_t \omega_t = \frac{1}{\mathrm{Vol}_t(U)} \int_U K_t u_t \,\mathrm{d}x\,\mathrm{d}y.$$

The *normalized Ricci flow equation* on $U$ is

$$\partial_t g_t = (K_{g_t} - \rho_t) g_t. \tag{6}$$

The left hand side is $\partial_t(u_t g_0) = (\partial_t u_t) g_0$ and the right hand side is $(\frac{1}{u_t}\Delta \log u_t - \rho_t) u_t g_0$, so the above equation simplifies to

$$\partial_t u_t = \left(\frac{1}{u_t}\Delta \log u_t - \rho_t\right) u_t. \tag{7}$$

[1] It is commonplace to use $\mathrm{d}x^2$ to denote $(\mathrm{d}x)^2$; this should not be confused with $\mathrm{d}(x^2)$.
[2] The $t$ subscript does not mean derivative. We'll stick with $\partial$ notation for derivatives.

Now let $D \subseteq R^2$ be the unit disk and let $S$ be the boundary. Given an initial weight function $u_0$ and boundary function $u_t\big|_S = f$ for all $t$, we can perform Ricci flow on the disk. If we remove the term $\rho_t$ we get the unnormalized Ricci flow $\partial_t u_t = \Delta \log u_t$, and by fixing a constant value $\rho$ we obtain the Yamabe flow which can flow the metric to one of constant curvature $\rho$, meaning $K_t$ approaches a constant function $\rho$. However, both unnormalized Ricci flow and Yamabe flow have a tendency to diverge under certain conditions.

## Tasks

**Task 1:** Let the initial condition be $u_0 = 1$ and let the boundary condition be $f = 1$. Set $\rho = 1$. Perform Yamabe flow to approximate the uniformization metric with curvature $\rho$. Then try various other values of $\rho$.

**Task 2:** Attempt a similar Yamabe flow when the boundary condition is not radially symmetric.

**Task 3:** Once we solve Task 1, we can plot geodesics for various $t$ values, each with the same initial point and velocity.

**Task 4:** The solution to Task 1 should converge to the Fubini-Study metric, which has weight function $u = \frac{1}{(1+|z|^2)^2}$. This is the behavior on the "south pole chart" of the sphere. Using the mapping $w = \frac{1}{z}$ from the south pole chart to the north pole chart to glue together two copies of the initial conditions from Task 1, perform *normalized Ricci flow* on the whole sphere.